

## ASSIGNMENT : 6 – OPERATING SYSTEM INSTALLATION

GUIDED BY : VIPUL MISTRI

**Q.1 Download the latest Ubuntu Desktop ISO image and create a bootable USB drive using Rufus or Balena Etcher.**

**ANS :** Here is a step-by-step guide to download the latest **Ubuntu Desktop ISO image** and write it to a USB drive using either **Rufus** (Windows) or **Balena Etcher** (Windows, macOS, Linux).

### Step 1: Download the Ubuntu ISO Image

1. Go to the official Ubuntu download page: [ubuntu.com/download/desktop](https://ubuntu.com/download/desktop).
2. Download the latest **Ubuntu Desktop LTS** version (e.g., **Ubuntu 26.04.1 LTS** or **Ubuntu 24.04 LTS**).
3. Save the .iso file to your computer.

### Step 2: Prepare Your USB Flash Drive

- Insert a USB drive (**minimum 8 GB capacity**) into your computer.
- **Important:** Back up any sensitive files on the drive—creating a bootable USB will erase all existing data on the flash drive.

### Step 3: Flash the ISO to USB

Choose one of the two tools below:

#### Option A: Using Rufus (Windows Only)

1. Download and open [Rufus](#) (Portable or Installer version).
2. **Device:** Select your connected USB drive.
3. **Boot selection:** Click **SELECT** and browse to choose the downloaded Ubuntu .iso file.
4. **Partition scheme:**
  - Select **GPT** (for modern UEFI systems).
  - Target system will default to **UEFI (non CSM)**.
5. **Volume label:** Leave as default (or rename it to UBUNTU).
6. Click **START**.
7. If prompted to download additional ISOLINUX files or write in **ISO Image mode vs. DD Image mode**, select **ISO Image mode (Recommended)** and click **OK**.

## ASSIGNMENT : 6 – OPERATING SYSTEM INSTALLATION

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8. Confirm the data wipe prompt by clicking **OK**. Wait for the progress bar to reach 100% and show **READY**.

### Option B: Using Balena Etcher (Windows, macOS, Linux)

1. Download and install [balenaEtcher](#).
2. Launch Balena Etcher and click **Flash from file**. Select the downloaded Ubuntu .iso file.
3. Click **Select target** and select your USB flash drive.
4. Click **Flash!** (you may be prompted to enter your administrator/system password).
5. Wait for the flashing process and the automatic verification step to complete.

### Step 4: Boot from the USB Drive

1. Safely eject and re-plug your newly created bootable USB drive into the target computer.
2. Restart the computer and immediately spam the **Boot Menu Key** (commonly **F12**, **F11**, **F8**, **Del**, or **Esc** depending on your motherboard/laptop brand).
3. Select your USB flash drive from the boot list to launch the **Ubuntu Live installer**.

## ASSIGNMENT : 6 – OPERATING SYSTEM INSTALLATION

GUIDED BY : VIPUL MISTRI

**Q.2 Install Ubuntu Desktop on a virtual machine using VirtualBox or VMware Player, and take a screenshot of the desktop after successful installation.**

**ANS :** Here is the step-by-step guide to installing **Ubuntu Desktop** on a Virtual Machine using **Oracle VM VirtualBox**, followed by an ASCII diagram representing your post-installation desktop interface.

### Step-by-Step Installation Guide (VirtualBox)

#### 1. Download Required Software

- **VirtualBox:** Download and install the latest version from [virtualbox.org](https://www.virtualbox.org).
- **Ubuntu ISO:** Download the latest ISO (e.g., **Ubuntu 26.04 LTS** or **24.04 LTS**) from [ubuntu.com/download/desktop](https://ubuntu.com/download/desktop).

#### 2. Create the Virtual Machine

1. Open VirtualBox and click **New**.
2. Set the VM configurations:
  - **Name:** Ubuntu-VM
  - **Folder:** Choose your storage path.
  - **ISO Image:** Browse and select your downloaded Ubuntu .iso file.
  - **Type:** Linux
  - **Version:** Ubuntu (64-bit)
3. Check **Skip Unattended Installation** (recommended if you want to configure settings manually during setup) and click **Next**.
4. **Hardware Allocation:**
  - **RAM:** Minimum 4096 MB (4 GB) or higher.
  - **Processors:** At least 2 CPUs.
5. **Virtual Hard Disk:**
  - Select **Create a Virtual Hard Disk Now**.
  - Size: Allocate at least **25 GB** to **50 GB**
6. Click **Finish**.

#### 3. Adjust VM Settings

## ASSIGNMENT : 6 – OPERATING SYSTEM INSTALLATION

GUIDED BY : VIPUL MISTRI

1. Select your new VM and click **Settings**.
2. Go to **Display** → **Screen**:
  - Drag **Video Memory** to 128 MB.
  - Check **Enable 3D Acceleration**.
3. Go to **Storage**: Ensure the downloaded Ubuntu ISO is attached under the Controller: IDE / SATA optical drive
4. Click **OK**.

### 4. Run the Ubuntu OS Installation

1. Click **Start** to boot the virtual machine.
2. Select **Try or Install Ubuntu** from the GRUB boot menu.
3. Choose your **Language** and click **Install Ubuntu**.
4. Configure Keyboard Layout, Network, and select **Interactive Installation**.
5. Choose **Default Selection** (or **Expanded Selection** if you want full media packages pre-installed).
6. Under **Installation Type**, select **Erase disk and install Ubuntu** (this only affects the virtual disk created in Step 2).
7. Create your user credentials:
  - Enter **Your Name**, **Computer Name**, **Username**, and a secure **Password**.
8. Click **Install**. Wait for the installation process to copy files and finish.
9. Click **Restart Now**. When prompted to remove the installation medium, press **ENTER**.

### 5. Post-Installation Optimization

1. Log into your fresh Ubuntu desktop.
2. Go to the VirtualBox top menu: **Devices** → **Insert Guest Additions CD Image...**
3. Run the installer inside Ubuntu to enable full-screen auto-resizing, shared clipboard, and smooth mouse integration.

**Q.3 During installation, set up a user account named 'cyberstudent' with a strong password (at least 8 characters, mix of letters, numbers, and symbols).**

## ASSIGNMENT : 6 – OPERATING SYSTEM INSTALLATION

GUIDED BY : VIPUL MISTRI

**ANS :** To set up the cyberstudent account during or after your Ubuntu Desktop installation, follow these details:

### 1. Account Details

- **Username:** cyberstudent
- **Password Requirements:** At least 8 characters containing a mix of uppercase/lowercase letters, numbers, and special symbols (e.g., Cyber@2026!Pass).

### 2. Setting Up During Installation

When you reach the **"Who are you?"** setup screen in the Ubuntu installer:

1. **Your name:** Cyber Student
2. **Your computer's name:** cyberstudent-vm (or default)
3. **Pick a username:** cyberstudent
4. **Choose a password:** Enter your strong password (e.g., Cyber@2026!Pass).
5. **Confirm your password:** Re-enter the exact same password.
6. Select **"Require my password to log in"** and click **Continue**.

### 3. Creating the Account via Terminal (Post-Installation)

If you already completed the installation with a different username and want to add cyberstudent with root/sudo privileges:

1. Open Terminal (Ctrl + Alt + T).
2. Run the command to create the user:

Bash

```
sudo adduser cyberstudent
```

3. Enter your strong password when prompted and complete the user details.
4. Grant the new account administrative (sudo) privileges:

Bash

```
sudo usermod -aG sudo cyberstudent
```

**Q.4 After installing Ubuntu, open the terminal and run the command 'uname -a' to display OS details. Save the output to a text file named os-info.txt.**

## ASSIGNMENT : 6 – OPERATING SYSTEM INSTALLATION

GUIDED BY : VIPUL MISTRI

**ANS :** Here are the step-by-step commands to run `uname -a` and save the output directly to `os-info.txt`:

1. Open the terminal by pressing `Ctrl + Alt + T`.
2. Run the following command to execute `uname -a` and redirect its output into `os-info.txt`:

Bash

```
uname -a > os-info.txt
```

3. To verify that the file was created and contains your system details, run:

Bash

```
cat os-info.txt
```